Sol. (*i*) Given expression $= \left(\dfrac{a^3 - b^3}{a^2 + ab + b^2}\right)$, where $a = 2.75$, $b = 2.25$

$= (a - b) = (2.75 - 2.25) = 0.5.$

(*ii*) Given expression $= \left(\dfrac{a^3 + b^3}{a^2 - ab + b^2}\right)$, where $a = 0.0347$, $b = 0.9653$

$= (a + b) = (0.0347 + 0.9653) = 1.$

Ex. 25. ***Find the value of:*** $\dfrac{(0.01)^2 + (0.22)^2 + (0.333)^2}{(0.001)^2 + (0.022)^2 + (0.0333)^2}$.

Sol. Given expression $= \dfrac{a^2 + b^2 + c^2}{\left(\dfrac{a}{10}\right)^2 + \left(\dfrac{b}{10}\right)^2 + \left(\dfrac{c}{10}\right)^2}$, where $a = 0.01$, $b = 0.022$, $c = 0.0333$ $= \dfrac{100\,(a^2 + b^2 + c^2)}{(a^2 + b^2 + c^2)} = 100.$

EXERCISE

(OBJECTIVE TYPE QUESTIONS)

Directions: ***Mark (✓) against the correct answer:***

1. The value of $\dfrac{42}{10000}$ in decimal fraction is **(L.I.C.A.D.O., 2008)**

(*a*) .0042 (*b*) .00042
(*c*) .0420 (*d*) 42000

2. The fraction $101\dfrac{27}{100000}$ in decimal form is

(*a*) .01027 (*b*) .10127
(*c*) 101.00027 (*d*) 101.000027

3. When .36 is written in simplest fractional form, the sum of the numerator and the denominator is

(*a*) 15 (*b*) 34
(*c*) 114 (*d*) 135

4. The place value of 9 in 0.06945 is **(L.I.C.A.D.O., 2008)**

(*a*) 9 (*b*) $\dfrac{9}{10}$
(*c*) $\dfrac{9}{100}$ (*d*) $\dfrac{9}{1000}$

5. What decimal of an hour is a second?

(*a*) .0025 (*b*) .0256
(*c*) .00027 (*d*) .000126

6. If $47.2506 = 4A + \dfrac{7}{B} + 2C + \dfrac{5}{D} + 6E$, then the value of 5A + 3B + 6C + D + 3E is

(*a*) 53.6003 (*b*) 53.6031
(*c*) 153.6003 (*d*) 213.0003

7. Express $\dfrac{1999}{2111}$ in decimal. **(R.R.B., 2006)**

(*a*) 0.893 (*b*) 0.904
(*c*) 0.946 (*d*) 0.986

8. How many times is 0.1 greater than 0.01?

(*a*) 1 time (*b*) 10 times
(*c*) 100 times (*d*) 1000 times

9. Out of the fractions $\dfrac{3}{7}, \dfrac{4}{11}, \dfrac{5}{9}, \dfrac{6}{13}, \dfrac{7}{12}$, which is the second largest? **(Bank Recruitment, 2010)**

(*a*) $\dfrac{3}{7}$ (*b*) $\dfrac{4}{11}$
(*c*) $\dfrac{5}{9}$ (*d*) $\dfrac{6}{13}$
(*e*) $\dfrac{7}{12}$

10. Which of the following has fractions in ascending order?

(*a*) $\dfrac{2}{3}, \dfrac{3}{5}, \dfrac{7}{9}, \dfrac{9}{11}, \dfrac{8}{9}$ (*b*) $\dfrac{3}{5}, \dfrac{2}{3}, \dfrac{9}{11}, \dfrac{7}{9}, \dfrac{8}{9}$
(*c*) $\dfrac{3}{5}, \dfrac{2}{3}, \dfrac{7}{9}, \dfrac{9}{11}, \dfrac{8}{9}$ (*d*) $\dfrac{8}{9}, \dfrac{9}{11}, \dfrac{7}{9}, \dfrac{2}{3}, \dfrac{3}{5}$
(*e*) $\dfrac{8}{9}, \dfrac{9}{11}, \dfrac{7}{9}, \dfrac{3}{5}, \dfrac{2}{3}$

11. Which of the following are in descending order of their values?

(*a*) $\dfrac{5}{9}, \dfrac{7}{11}, \dfrac{8}{15}, \dfrac{11}{17}$ (*b*) $\dfrac{5}{9}, \dfrac{8}{15}, \dfrac{11}{17}, \dfrac{7}{11}$
(*c*) $\dfrac{11}{17}, \dfrac{7}{11}, \dfrac{8}{15}, \dfrac{5}{9}$ (*d*) $\dfrac{11}{17}, \dfrac{7}{11}, \dfrac{5}{9}, \dfrac{8}{15}$

12. What is the difference between the biggest and the smallest fraction among $\dfrac{2}{3}, \dfrac{3}{4}, \dfrac{4}{5}$ and $\dfrac{5}{6}$? **(S.S.C., 2006)**

(*a*) $\dfrac{1}{6}$ (*b*) $\dfrac{1}{12}$
(*c*) $\dfrac{1}{20}$ (*d*) $\dfrac{1}{30}$